

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres
One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points

★★ CORE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

- (i) Show that $I = (x) \cap (y, z)$.
- (ii) Determine the irreducible components and their generic points.
- (iii) Is X reduced, connected, irreducible, or integral?

Exercise 2. The function field of the cusp

★★ CORE

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral, compute its function field, and deduce its dimension.

Exercise 3. Chains and dimensions

★ CORE

- (i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.
- (ii) Give a chain of length n in $\mathbb{A}_{\mathbb{k}}^n$.
- (iii) Determine the dimension of

$$\operatorname{Spec}(\mathbb{k}[x, y, z]/(xy, xz))$$

from its components.

Exercise 4. Noether normalisation and dimension

★★ CORE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x^3).$$

- (i) Show that B is finite over $\mathbb{k}[x]$.
- (ii) Use this to recover $\dim(B) = 1$.
- (iii) Compare with the computation from the function field.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line

★ CORE

Let

$$\pi : \mathbb{A}_{\mathbb{C}}^2 = \operatorname{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}_{\mathbb{C}}^1 = \operatorname{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t - a)$ and its fibre over the generic point (0) . Explain why the latter is $\operatorname{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Exercise 6. A fibre product larger than the set-theoretic one

★★ CHOICE

Let $X = \operatorname{Spec}(\mathbb{C})$ and $S = \operatorname{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Exercise 7. The power map and its fibres

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}_{\mathbb{k}}^1 \longrightarrow \mathbb{A}_{\mathbb{k}}^1, \quad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Exercise 8. A family whose special fibre becomes reducible

★★ **CORE**

Let

$$X = \operatorname{Spec}(\mathbb{k}[x, y, t]/(xy - t)) \longrightarrow \mathbb{A}_t^1.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Exercise 9. A jumping fibre

★★ **CORE**

Consider

$$f : \mathbb{A}_{\mathbb{k}}^2 \longrightarrow \mathbb{A}_{\mathbb{k}}^2, \quad (x, y) \longmapsto (x, xy).$$

Compute the fibre over (a, b) and show that the generic fibre has dimension 0, while the fibre over $(0, 0)$ has dimension 1.

Exercise 10. The generic fibre formula in examples

★★ **CHOICE**

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}_{\mathbb{k}}^{m+n} \rightarrow \mathbb{A}_{\mathbb{k}}^m$.

Alternative route: separatedness and properness

Exercise 11. The valuative criterion: affine versus projective space

★★ **CORE**

Let $R = \mathbb{k}[[\pi]]$ and $K = \mathbb{k}((\pi))$.

- (i) Show that R is a valuation ring and describe $\operatorname{Spec}(R)$.
- (ii) Use the K -point $\pi^{-1} \in \mathbb{A}_{\mathbb{k}}^1(K)$ to prove that $\mathbb{A}_{\mathbb{k}}^1$ is not proper over $\mathbb{k}$.
- (iii) Given $[a_0 : \cdots : a_n] \in \mathbb{P}_{\mathbb{k}}^n(K)$, multiply the coordinates by a common power of π so that all lie in R and at least one is a unit. Deduce that the point extends to an R -point of $\mathbb{P}_{\mathbb{k}}^n$.
- (iv) Explain where separatedness supplies uniqueness and conclude, using the valuative criterion, that $\mathbb{P}_{\mathbb{k}}^n$ is proper.

Exercise 12. Global functions on a proper integral variety

★★★ **CORE**

Let X be an integral proper scheme of finite type over a field $\mathbb{k}$, and put $B = \Gamma(\mathcal{O}_X)$. For $b \in B$:

- (i) construct the corresponding morphism $g_b : X \rightarrow \mathbb{A}_{\mathbb{k}}^1$;
- (ii) put $C = \operatorname{im}(\mathbb{k}[t] \rightarrow B)$ and $Z = \operatorname{Spec}(C)$. Show that g_b factors as a dominant morphism $X \rightarrow Z$ followed by a closed immersion $Z \hookrightarrow \mathbb{A}_{\mathbb{k}}^1$;
- (iii) use properness to show that Z is proper over $\mathbb{k}$, then use the fact that an affine proper morphism is finite to prove that b is algebraic over $\mathbb{k}$;
- (iv) let $\mathbb{k}_X$ be the algebraic closure of $\mathbb{k}$ inside the finitely generated field $K(X)$. Prove that

$$B \subseteq \mathbb{k}_X,$$

and conclude that B is finite-dimensional over $\mathbb{k}$. What happens when $\mathbb{k}$ is algebraically closed?

Optional outreach exercises: Hilbert schemes of points

Exercise 13. $\operatorname{Hilb}^n(\mathbb{A}^1)$

★★ **CHOICE**

Assume that $\mathbb{k}$ is algebraically closed. Show that every ideal $I \subset \mathbb{k}[x]$ with $\dim_{\mathbb{k}} \mathbb{k}[x]/I = n$ is generated by a unique monic polynomial of degree n . Deduce

$$\operatorname{Hilb}^n(\mathbb{A}_{\mathbb{k}}^1) \simeq \mathbb{A}_{\mathbb{k}}^n$$

and explain how factorisation records support and multiplicities.

Exercise 14. Commuting matrices and a cyclic vector

★★★ **OPTIONAL**

Let V be n -dimensional and let (X, Y, v) satisfy $[X, Y] = 0$ and $\mathbb{k}[X, Y]v = V$. Show that

$$I = \{f \in \mathbb{k}[x, y] \mid f(X, Y)v = 0\}$$

is an ideal and $\mathbb{k}[x, y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.